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PAPER

Analogies between phase transitions in potential games and quantum phase transitions

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Keywords: game theory, quantum games, statistical mechanics

Abstract

Potential games at population level has a very natural analogy with statistical mechanical systems. Here we show that there are clear analogies between quantum phase transitions at zero temperature and phase transitions in potential games being played by fully rational players. Such phase transitions are brought about by tuning parameters which change the payoff matrix either directly (as in classical games) or indirectly through continuous change in strategies (as in quantum games). The phase transitions take the system from one Nash equilibrium to another; these Nash equilibria (NE) are, in a sense, refined as only the ones that correspond to global maxima of the potential are selected in the thermodynamic limit (infinite number of players). We observe that the types of the phase transitions depend on the states involved in the transition process: while transitions involving two symmetric NE are discontinuous, the transitions between a symmetric and an antisymmetric NE are continuous.

1. Introduction

The famous exclamatory remark (in German) of von Neumann to Morgenstern [1]—the two pioneers of game theory—'*Ja hat denn das niemand gesehen?*' (Google's English translation: 'Yes, did not anyone see that?') is a lingering testimony of fascinating hidden analogies between thermodynamics and game theory; after all, the thermodynamical concepts of heat and temperature laid the foundations of axiomatic treatment [2] of utility and rationality, and thence, the game theory.

Such analogies have only flourished over the years: another important idea—phase transition (either equilibrium or non-equilibrium)—in thermodynamics and statistical mechanics has been well-explored [3–9] in both classical and evolutionary games. It is not surprising that various collective configurations of a population of players playing games would show up as parameters of the strategic interaction between players change. For example, noise is realistically present in every real system—while temperature characterizes it in thermodynamic system, intensity of rationality does so in game theoretic system; the less rational a player, the more seemingly random choice of strategy is by the player. Thus, as varying temperature leads to some common phase transitions (like liquid to gas and paramagnet to ferromagnet) in nature, varying intensity of rationality may lead to a phase where the entire population collectively plays a Nash equilibrium (NE) strategy—a strategy that when deviated from unilaterally does not fetch any benefit to the deviating player.

This paper looks to add a new chapter to such analogies. A highly fertile field of research in physics is that of quantum phase transition [10–16], theoretically realized at zero temperature. Naturally, one should look for an analogy of such phase transitions in the context of games in the limit of fully rational ('zero-temperature') players. To this end, it is pragmatic to follow the mathematical scheme used

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in physics: the language of Hamiltonian and partition function. This is best realized in potential games, where a single global function captures the incentives of all players-much like a conservative potential in physics-making them particularly amenable to tools from statistical mechanics.

The concept of potential games originated with the seminal work of Monderer and Shapley [17]. They demonstrated that multi-agent interactions can be characterized by a single potential function that encapsulates the collective incentives of all players. In such games, when any player does unilateral strategy-change improving their individual payoff, it results in an equivalent corresponding change in the potential function, thereby linking individual rationality with the global dynamics of the system. This property makes potential games particularly useful across economics, computer science, and engineering, where systems of self-interested agents tend to converge naturally to stable equilibria (Nash equilibria (NE)) without requiring centralized coordination [18–21]. In contrast, the literature on quantum potential games remains at a very nascent stage compared to that of classical potential games or general quantum games. A limited number of studies in quantum game theory have highlighted the applicability of these hybrid frameworks for distributed decision-making in engineered and quantum-enabled systems [22, 23]. Thus, needless to say, quantum potential games may represent a promising intersection of classical game-theoretic structures and quantum strategy spaces, offering new modeling tools for analyzing equilibrium behavior in emerging quantum technologies.

The paper is organized as follows: in section 2, we introduce the Hamiltonian and the partition function formalism of potential games. Section 3 analyzes the corresponding phase transition behavior. In section 4, we extend the framework to quantum potential games and examine their phase transition characteristics, while section 5 discusses possible indicators of quantum-like phase transitions. Finally, section 6 presents the overall discussion and conclusions.

2. Hamiltonian formalism of potential games

In the context of one-shot two-player bimatrix game, a potential game [17, 24–26] is the one where a unique (up to an additive constant) global potential V matrix exists such that

$$\begin{aligned} \mathsf{V}_{kj} - \mathsf{V}_{ij} &= \Pi^A_{kj} - \Pi^A_{ij}, \\ \mathsf{V}_{il} - \mathsf{V}_{ij} &= \Pi^B_{il} - \Pi^B_{ij}; \end{aligned} \tag{1}$$

where Π^A and Π^B are the payoff matrices of player A and player B, respectively. The subscripts in the equation above refer to the elements of the corresponding matrix: e.g. Π^A_{kj} is the payoff of player A when she plays action s_k (out of, say, n_A possible actions available in her strategy set $\mathcal{S}_A$) against action s_j (out of, say, n_B possible actions available to her in her strategy set $\mathcal{S}_B$). Ergo, $i,k \in \{1,2,\cdots,n_A\}$ and $j,l \in \{1,2,\cdots,n_B\}$. Thus notationally, $\mathsf{V}_{il} = \mathsf{V}(s_i,s_l)$. In the paper, we shall stick with $\mathcal{S}_A = \mathcal{S}_B = \mathcal{S} \implies n_A = n_B = n$. Such a potential approach simplifies the analysis of the equilibrium: the strategy profile for which V_{ij} achieves its maximum (with respect to single player deviations) is a preferred rational outcome—an NE; if the maximum is global then the corresponding NE strategy profile may aptly be called the *most probable one*—justification of this terminology may be sought in the methodology of statistical mechanics elaborated below. Interestingly, for 2×2 games, the global maximum corresponds to the risk-dominant NE [27–29].

Interestingly, the long-established correspondence [3, 8] between the potential in a game-theoretic problem and the Hamiltonian in a many-body system permits us to construct a common viewpoint to observe the ground state or equilibrium property of a system and its variation with system parameters. Allowing Greek superscripts to tag locations (nodes) of agents on a graph, the Hamiltonian of a many-agent interaction may simply be defined as

$$H = \sum_{\langle \mu\nu \rangle} \mathsf{V}(s^\mu, s^\nu)\, G_{\mu\nu}, \quad s^\mu, s^\nu \in \mathcal{S}; \tag{2}$$

where adjacency matrix, $G_{\mu\nu} = 1$ if the agents/players, indexed by μ and ν, interact; otherwise, $G_{\mu\nu} = 0$. The notation $\langle \mu\nu \rangle$ mean that μ and ν interact. Viewing the system as a statistical mechanical one, when the system size is large, at *equilibrium*, one may also define a partition function [30, 31]:

$$Z(K) = \sum_{s^1=s_1}^{s_n} \sum_{s^2=s_1}^{s_n} \cdots \sum_{s^N=s_1}^{s_n} \exp\left(\frac{H}{K}\right). \tag{3}$$

This parameter K— a measure of deviation from full rationality [8, 27]—can be understood as an analogy of the temperature of a thermodynamic system. We note that while writing the partition function, we take $G_{\mu\nu} = 1 \forall \mu, \forall \nu \neq \mu$, to envisage an equilibrium canonical ensemble corresponding to a population (of a large size N) where every player interacts with every other player through a binary interactions—there are always $\binom{N}{2}$ pairwise interactions. Effectively, we consider $\binom{N}{2}$ pairwise interactions among N individuals, where each interaction is modeled as a two-player, n-strategy game. As evident from equation (3), the number of players N does not explicitly enter the payoff matrix, although it implicitly affects the partition function via the multiplicity of microstates. One may consider every possible assignment of strategies to players as a probable microstate of the ensemble. One may note that the form of our derived partition function aligns with the Gibbs distribution in the equilibrium state, provided we assume the Glauber dynamics (smoothened best response or Logit, in the game theory parlance) as the microscopic update rule for the macroscopic system [3].

In passing, we remark that while the conventional formulation of the partition function includes a negative sign in the argument of exponential (that is, $-H/K$), adopting a positive sign (see equation (3)) leaves the underlying physics unchanged and can be interpreted as corresponding to a negative temperature or negative potential energy. This convention reflects our system's evolution toward a state of maximum potential (or payoff), consistent with the maximization framework of evolutionary game theory and in agreement with earlier studies [32–34].

3. Phase transitions

Due to the presence of the exponential function, all n^N summands in the partition function will contribute to the sum when noise K is non-zero; in the limit of zero noise or $K \to 0$, the system can only be found at the highest potential. Thus, any shift from the microstate corresponding to the argmax set of the potential is qualitatively equivalent to a phase transition in the system. As the potential is uniquely mapped to the payoff (see equation (1)), the collective payoff of the system can be treated as an order parameter for such phase transitions. The quantitative effect of such phase transitions will be reflected either in the discontinuous change of the internal energy or its derivative at the critical point. The equivalent of the internal energy of the system are given as

$$\langle E \rangle = -K^2 \frac{\partial}{\partial K} \ln Z(K) . \tag{4}$$

The following two subsections analyze the case of two-player two-strategy symmetric game. Such games are known to be potential games. The payoff elements and potential matrix are described in equations (7) and (8).

3.1. Ferromagnetic-to-ferromagnetic phase transition

Example of a phase would be where every individual plays the same strategy (symmetric equilibrium) in the zero noise limit; it is analogous to the *ferromagnetic phase* in magnetic systems. In figure 1 we depict these phases, respectively phase I and III, corresponding to two possible maximum potential elements, V_{CC} and V_{DD} respectively. Without any loss of generality, let V_{mm} and $\mathsf{V}_{m'm''}$ (where $(m', m'') \neq (m, m)$) be the largest and the second largest elements, respectively, in the potential matrix. The population state corresponding to the second highest energy configuration is always the case where $(N-1)$ players play the strategy m and only one plays m' (see appendix A). In the limit of small noise (almost rational players), the partition function in equation (3) turns out to be

$$\begin{aligned} Z(K) &= \mathrm{e}^{\frac{1}{K}\binom{N}{2} V_{mm}} + N \mathrm{e}^{\frac{1}{K}(N-1) V_{mm'}} \mathrm{e}^{\frac{1}{K}\binom{N-1}{2} V_{mm}} + \cdots \\ &= \mathrm{e}^{\binom{N}{2}\frac{1}{K}\mathsf{V}_{mm}} \left[1 + N \mathrm{e}^{-\frac{N-1}{K}\Delta} \right] + \cdots , \end{aligned} \tag{5}$$

where, $\Delta \equiv \mathsf{V}_{mm} - \mathsf{V}_{mm'}$ is the energy gap between the top two most probable states and we assume that $\Delta > 0$ (non-degenerate eigenstates). Therefore, using equation (4)

$$\langle E \rangle = \binom{N}{2} \mathsf{V}_{mm} - \frac{N(N-1)\Delta}{\mathrm{e}^{\frac{(N-1)\Delta}{K}} + N}, \tag{6}$$

in the limit $K \to 0$.

A phase transition due to tuning of parameters, other than the temperature-like K, while the system stays in a symmetric equilibrium is most simply comprehended in the context of two-player two-strategy

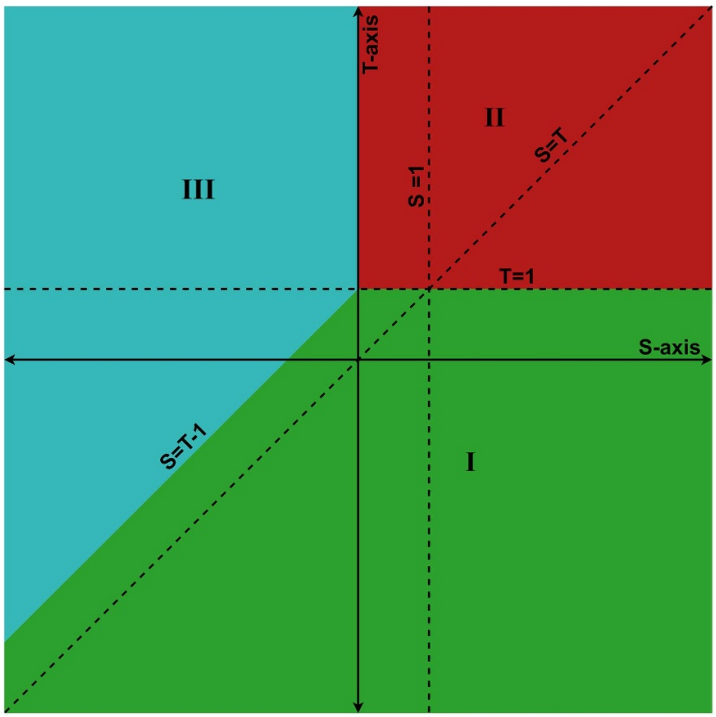

Figure 1. A schematic presentation of twelve ordinal game classes demarcated by the lines: $S=0$, $T=0$, $S=1$, $T=1$, and $S=T$. On the basis of the strategies, corresponding to the highest potential element, there can be three distinct classes: the green region (region (I) with the strategy profile (C, C), the red region (region II) with the strategy profiles (C, D) and (D, C), and cyan region (region III) with the strategy profile (D, D) correspond to the highest potential (see [35] for detailed description of the game classes).

symmetric games which are always potential games [17]. A 2×2 payoff bimatrix is commonly represented in the following form:

$$\begin{array}{c|c|c|} & \mathrm{C} & \mathrm{D} \\ \hline \mathrm{C} & (R,R) & (S,T) \\ \hline \mathrm{D} & (T,S) & (P,P) \\ \hline \end{array}. \tag{7}$$

Here, the payoff elements R, S, T, and P, are popularly known as the Reward, Sucker's payoff, Temptation, and Punishment, respectively [36]; and the actions/strategies C and D are termed Cooperation and Defection, respectively. Therefore, R and P correspond to the payoffs for mutual cooperation and mutual defection, respectively. Without any loss of generality, to make our analysis tractable, we set $R=1$ and $P=0$—which amounts to performing a positive affine transformation on the payoff elements [35]—for the classical games. The resulting payoff bimatrix corresponds to the following potential matrix:

$$\mathsf{V}=\begin{pmatrix} 0 & T-1 \\ T-1 & T-S-1 \end{pmatrix}. \tag{8}$$

Owing to the existence of a common $\mathbb{Z}_2$ symmetry, the corresponding partition function [31, 37, 38] is shown to be analogous to that of a long-ranged classical Ising model with N spins in a transverse magnetic field: the interaction strength J and the magnetic field strength h in the Ising model is analogous to the payoff dependent functions $(1-S-T)/4$ and $(T-S-1)/2$, respectively, in the game-theoretic setup [3].

Even at zero noise, a ferromagnetic-to-ferromagnetic phase transition is clearly evident when the potential maximum shifts between symmetric strategy profiles, such as moving from (D,D) to (C,C) or vice-versa. A comparison of the four elements of the potential matrix reveals (D,D) is the most probable NE when $T\leqslant 1$ and $S>T-1$; and (C,C) is the most probable NE when $T\leqslant 1$ and $S<T-1$. Thus, as per figure 1, the phase transition is possible when parameters change to take one from cyan region to green region across the line $S=T-1$. In terms of the games, such transitions are possible either

between a Prisoner's Dilemma (PD) game and a Harmony game or between two appropriate coordination games.

A transition parameter for this case can be defined as $\lambda \equiv \mathsf{V}(D,D) - \mathsf{V}(C,C) = T - S - 1$. We note that such transition parameters are defined so that the zero value of λ serves as a boundary between two game classes (see figure 1), each characterized by a distinct set of NE. In the limit of zero noise and in the thermodynamic limit, the energies of the system in these two ferromagnetic phases can be shown to be of the following form:

$$\langle E \rangle_I = 0, \tag{9a}$$

$$\langle E \rangle_{III} = \frac{N^2}{2}(T - S - 1). \tag{9b}$$

One may note that the energies of these two phases are continuous on the phase boundaries. However, the same is not true for the first derivative of the energies:

$$\delta \left| \frac{\partial \langle E \rangle}{\partial \lambda} \right| \equiv \left| \frac{\partial \langle E \rangle_I}{\partial \lambda}\bigg|_{\lambda=0^+} - \frac{\partial \langle E \rangle_{III}}{\partial \lambda}\bigg|_{\lambda=0^-} \right| = \frac{N^2}{2}, \tag{10}$$

showing a discontinuous jump in the derivative of average energy across the phase transition. It points towards a very interesting result: ferromagnetic-to-ferromagnetic transitions are discontinuous transitions. For clarity, we have elaborated on test examples of this type of transition in appendix B.

3.2. Antiferromagnetic-to-ferromagnetic transition

A completely different kind of phase appears when there is a transition from the PD, Harmony Game or coordination game to the anti-coordination game ($S > 0$ and $T > 1$), characterized by coexistence of asymmetric NE and mixed NE. As per figure 1, we are talking about the transition between cyan or green region and red region. This phase (in red region) reminds one of the antiferromagnetic phase (all spins are not aligned in the same direction): the full population cannot play the same strategy in the zero noise limit owing to the restrictions imposed by the Hamiltonian. Here, the maximum potential element $\mathsf{V}_{mm'}$ is asymmetric in the strategy, i.e. $m \neq m'$. If α and $(N-\alpha)$ are the numbers of agents with strategy s_m and $s_{m'}$, respectively, then with

$$V_{\text{tot}}(\alpha) = \frac{1}{2}\alpha(\alpha-1)\mathsf{V}_{mm} + \alpha(N-\alpha)\mathsf{V}_{mm'} + \frac{1}{2}(N-\alpha)(N-\alpha-1)\mathsf{V}_{m'm'}, \tag{11}$$

denoting the total energy of the system, the partition function of equation (3) becomes

$$Z(K) = \mathrm{e}^{\frac{V_{\text{tot}}(\alpha)}{K}}\left[1 + \mathrm{e}^{\frac{V_{\text{tot}}(\alpha\pm1) - V_{\text{tot}}(\alpha)}{K}}\right], \tag{12}$$

for infinitesimal noise, up to first order in $\tilde{\Delta} = V_{\text{tot}}(\alpha) - V_{\text{tot}}(\alpha \pm 1)$. Thus, for small noise, in this phase,

$$\langle E \rangle = V_{\text{tot}}(\alpha) - \frac{\tilde{\Delta}}{1 + \mathrm{e}^{\frac{\tilde{\Delta}}{K}}}. \tag{13}$$

An asymmetric strategy profile corresponding to the maximum potential matrix element means that the potential is maximized in each binary interaction given the involved players play different strategies. In a population as considered in our case, where the agents interact with each other in an unstructured geometry, the equilibrium state of the population has to be interpreted as a *geometrically frustrated state* [39–41]: many macroscopically equivalent degenerate microstates are possible. To give an example, let us consider a population of three player interacting with each other through 2×2 anti-coordination game; each pair of players should have opposite strategies to maximize the potential. But the obvious indeterminacy of strategy of the third player—who is supposed to interact with both the players simultaneously—gives rise to six-fold degenerate population states. The specific microstate realized at equilibrium starting from a non-equilibrium state depends on the underlying protocol of the strategy-update-rule.

A caveat is worth pointing out: understanding the asymmetric NE at the population level in anti-coordination games requires considering two populations with interspecific interactions. However, the statistical mechanical setup we are interested in in this paper allows only for a single-population interpretation of the asymmetric NE. Therefore, it not surprising that in any game featuring asymmetric potential maxima, particularly in set-ups involving more than two players, statistical mechanics tends

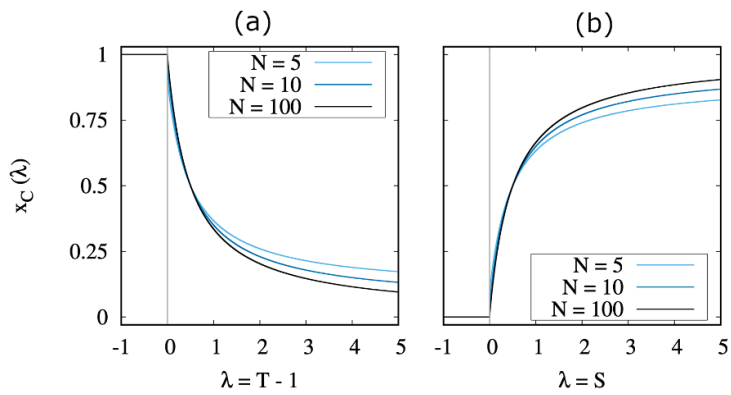

Figure 2. Change in cooperator-fraction (x_C) during antiferromagnetic-to-ferromagnetic transition: subfigure (a) depicts the transition from region I to region II whereas subfigure (b) depicts the same for region II to region III. Cyan, blue and black curves, respectively, correspond to $N = 5$, $N = 10$, and $N = 100$ sizes of the population.

to favor a polymorphic population that somewhat aligns with the concept of mixed Nash Equilibrium state, even if this does not directly correspond to the potential maxima present in a two-player scenario. We remark that a polymorphic population means coexistence of individuals using different strategies that persist stably over time. It is the population-level analogue of a mixed phase in physics—a dynamic equilibrium between competing 'species' or 'states' (strategies). Such an equilibrium, corresponding to maximization of V_{tot} (corresponding to the anti-coordination game; see equation (11)) with respect to x (where $x \equiv \alpha/N$ be the fraction of agents playing s_m), is given by

$$x = \frac{S + \frac{1}{2N}(T - S - 1)}{S + T - 1}. \tag{14}$$

Obviously, in the limit of infinite population, $x = S/(S + T - 1)$, which interestingly, is the mixed (symmetric) NE of the anti-coordination game [42]. This limit is understood as follows. Note first that the mixed NE is obtained under maximization of V_{tot}, irrespective of population size, if we take self-interaction into account, i.e. every player can also engage with itself—this technically means to replace '-1' in equation (11) by '0'. Therefore, in the thermodynamic limit, i.e. large N, the contribution from self-interaction becomes relatively negligible, allowing for the standard classical game theory outcome.

In the limit of zero noise and in the thermodynamic limit, the energy of the system in the antiferromagnetic phase is given by

$$\langle E \rangle_{II} = \frac{N^2}{2} \frac{(T-1)^2}{(T + S - 1)}. \tag{15}$$

One can note that the energy of this phase is continuous on the phase boundaries with the other two ferromagnetic phases. Interestingly, the same is true for the first derivative of the energies (see appendix A):

$$\begin{aligned} \left| \frac{\partial \langle E \rangle_I}{\partial \lambda}\bigg|_{\lambda=0^+} - \frac{\partial \langle E \rangle_{II}}{\partial \lambda}\bigg|_{\lambda=0^-} \right| &= 0, \\ \left| \frac{\partial \langle E \rangle_{II}}{\partial \lambda}\bigg|_{\lambda=0^+} - \frac{\partial \langle E \rangle_{III}}{\partial \lambda}\bigg|_{\lambda=0^-} \right| &= 0. \end{aligned} \tag{16}$$

Thus, we can see that both the antiferromagnetic-to-ferromagnetic phase transitions are continuous transitions. We also note that in line with equation (14), the aforementioned phase transitions would involve continuous change in cooperator-fraction which should depend on population size as well (see figure 2).

We present the summary of both the ferromagnetic-to-ferromagnetic and antiferromagnetic-to-ferromagnetic phase transitions in 2×2 games in table 1 and also illustrate them on K–λ space in figure 3 to highlight their striking visual resemblance with the quantum phase transitions found in true quantum systems.

Table 1. Characteristics of quantum-like phase transitions in 2×2 classical games. (The fidelity-like measure, $f(\lambda)$, is described later in section 5).

Phases involved	Shift in NE states	Transition parameter (λ)	$\delta\,\lvert\partial\langle E\rangle/\partial\lambda\rvert$	$f(\lambda=0)$	Type of transition
Phase I and phase III	All C to all D	$T-S-1$	$\frac{N^2}{2}$	0	Discontinuous
Phase II and phase III	Mixed state to all D	S	0	$1-\frac{1}{2N}$	Continuous
Phase I and phase II	All C to mixed state	$T-1$	0	$1-\frac{1}{2N}$	Continuous

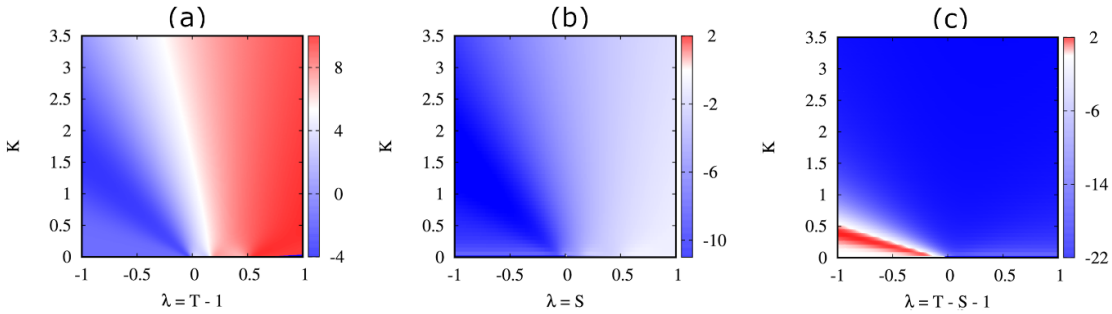

Figure 3. Illustrating quantum-like phase transitions in 2×2 classical games: the density plots depict the first derivative of energy with respect to the transition parameter on parameter space of noise strength (K) and transition parameter (λ). We have considered an all-to-all connected network of five players. (a), (b), and (c) depict transitions from region I to region II, region II to region III, and region I to region III, respectively. The resemblance of these plots with quantum phase transition plots (see figure 1.2(b) in textbook [10]) is intriguing.

4. Quantum potential games: phase transitions

We note that in a classical game to achieve the afore-discussed phase transitions, we had to tune the payoff elements leading to change in the ordinal class of the games at the phase transitions. A somewhat different scenario is when one of the players has access to a continuous strategy set and is capable of changing their strategy continuously to bring about phase transitions. One such kind of game with continuous strategy space is a quantum game [43–48].

The extension of the one-shot game into the quantum domain may be done using many existing protocols [43, 45, 49–52]. In our paper, we follow the Eisert–Wilkens–Lewenstein (EWL) protocol [45] that maps the classical strategies into a continuum set of quantum strategies available to the players. This extension helps us to arrive at the outcomes of the classical strategies C and D through manipulations of Hilbert state vectors $|0\rangle$ and $|1\rangle$ respectively. Generally, the state of the quantum game is defined as $|\psi\rangle \equiv \mathsf{J}(\varepsilon)|ab\rangle = \mathsf{J}(\varepsilon)|a\rangle_A \otimes |b\rangle_B$ where $a, b \in \{0, 1\}$ and the unitary entangling operator

$$\mathsf{J}(\varepsilon) \equiv \mathsf{1_A} \otimes \mathsf{1_B} \cos\frac{\varepsilon}{2} + \sqrt{-1}\,\mathsf{D_A} \otimes \mathsf{D_B} \sin\frac{\varepsilon}{2}. \tag{17}$$

(A and B tags to the two players, Alice and Bob, respectively.) Here, ε stands for the parameter characterizing the entanglement between the qubits, 1 is the identity operator and D is the standard qubit flip Y gate with a phase. The action of D on the states $|0\rangle$ and $|1\rangle$ takes it to its orthogonal counterpart $-|1\rangle$ and $|0\rangle$ respectively. The two-qubit entangling operator $\mathsf{J}(\varepsilon)$ acts on the joint Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$. While operating it forms an entangled two-qubit state from the pure states $|0\rangle_A$ and $|0\rangle_B$. The entanglement parameter, $\varepsilon \in [0, \pi/2]$, quantifies the entanglement of the initial states of the players.

Any quantum strategy is represented as a unitary operator belonging to a $SU(2)$ group rotation and is of the following form [53]

$$\mathsf{U} = \begin{pmatrix} \gamma & \delta \\ -\delta^* & \gamma^* \end{pmatrix} \tag{18}$$

with $|\gamma|^2 + |\delta|^2 = 1$; the elements are complex numbers and the asterisk denotes the complex conjugate. From now on, we will use the superscript A or B depending on whether Alice or Bob has played the strategy. In general, the initial game state is chosen as $|\psi_{\text{initial}}\rangle \equiv \mathsf{J}(\varepsilon)|00\rangle$. The simultaneous strategic action by Alice and Bob leads to a new final state $|\psi_{\text{final}}\rangle = \mathsf{J}^\dagger(\varepsilon)(\mathsf{U}^A \otimes \mathsf{U}^B)\mathsf{J}(\varepsilon)|00\rangle$. The payoff is given by $\$_A \equiv \sum_a \sum_b \Pi_{ab} |\langle ab|\psi_{\text{final}}\rangle|^2$ which basically weight the payoffs with the likeliness of the basis states. The extension from classical game to quantum game is justified as the following mapping holds: U as $\mathsf{1}$ and

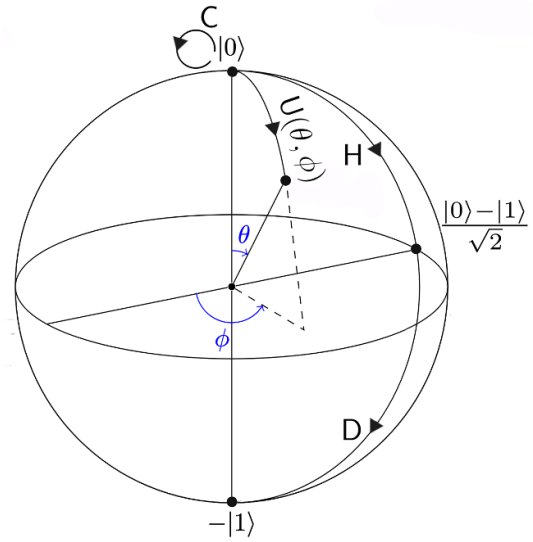

Figure 4. The actions of three quantum strategies Cooperation (C), Defection (D), Hadamard (H), and a generic strategy $\mathsf{U}(\theta,\phi)$ on the state $|0\rangle$ are depicted schematically on the Bloch sphere. All quantum strategies are 2×2 unitaries on the SU(2) Hilbert space and their actions on a pure state are represented by rotations on the surface of the Bloch sphere.

D—respectively representing cooperation and defection actions—the payoff is identical to that of a classical game, as expected. To make the study analytically tractable we focus on the standard two-parameter subspace of SU(2) where $\delta=\delta^*=\sin\frac{\theta}{2}$ and $\gamma=\mathrm{e}^{\mathrm{i}\phi}\cos\frac{\theta}{2}$; $\theta\in[0,\pi]$ and $\phi\in[0,\pi/2]$.

There is no guarantee that the potential structure exists for all quantum games; not much work is reported in the literature (cf in the context of evolutionary game dynamics [27, 54–57]). We restrict our attention to potential games because they enable the application of statistical mechanical prescriptions within the framework of game theory. The general condition for potentiality in quantum games is discussed in detail in appendix C (refer equation (C2)). We find (see appendix C) that one needs at least two quantum strategies along with a classical one to get a non-trivial quantum potential game. We prefer to choose the Hadamard gate H as it is one of the most well known quantum strategies studied in the literature [58]. To this end, let us choose Cooperation (C); the well-known Hadamard strategy, $\mathsf{H}\equiv(\theta=\frac{\pi}{2},\phi=0)$; and another quantum strategy, U whose θ and ϕ are such that we get a potential game structure (refer appendix C):

$$\sin\varepsilon=\frac{\tan\theta/2}{\sin\phi},\quad \forall\sin\varepsilon\neq 0. \tag{19}$$

The solution to the above equation forms a continuous line effectively parametrized by a single parameter belonging to $SU(2)$ subspace. From the form of the above transcendental equation and the allowed ranges of the trigonometric functions, it follows that the condition of potentiality imposes a restriction on θ. Consequently, the maximum allowed range of θ is $[0,\pi/2]$, which corresponds to the case of maximal entanglement with $\varepsilon=\pi/2$. For non-maximal entanglement, the upper bound of θ becomes strictly less than $\pi/2$. These strategies are pictorially shown in figure (4). The resulting 3×3 game, corresponding to the underlying classical game as defined in equation (7), possesses a potential matrix V whose elements are given by:

$$\begin{aligned}
&V_{11}=0,\quad V_{12}=V_{21}=\frac{T-R}{2},\quad V_{22}=\frac{1}{4}\left(P-3R-S+3T\right),\\
&V_{13}=V_{31}=(P-2R+T)\sin^2\left(\frac{\theta}{2}\right),\quad V_{23}=V_{32}=\frac{T-R}{2}+(2P-R-S)\sin^2\left(\frac{\theta}{2}\right),\\
&V_{33}=\frac{1}{2}\Bigg[-2R-S+T+(S-T)\cos\theta+2\cos^4\frac{\theta}{2}\left(R\cos^2\eta+\left(R\cos^2\varepsilon+P\sin^2\varepsilon\right)\sin^2\eta\right)\\
&\qquad+\sin^2\frac{\theta}{2}\left((P+S+T)+(-P+S+T)\cos\theta+2\left(2P-S-T\right)\sin\theta\cos\frac{\eta}{2}\right)\Bigg],\\
&\qquad\left[\text{where}\,\eta\equiv 2\sin^{-1}\left(\csc\varepsilon\tan\frac{\theta}{2}\right)\right].
\end{aligned} \tag{20}$$

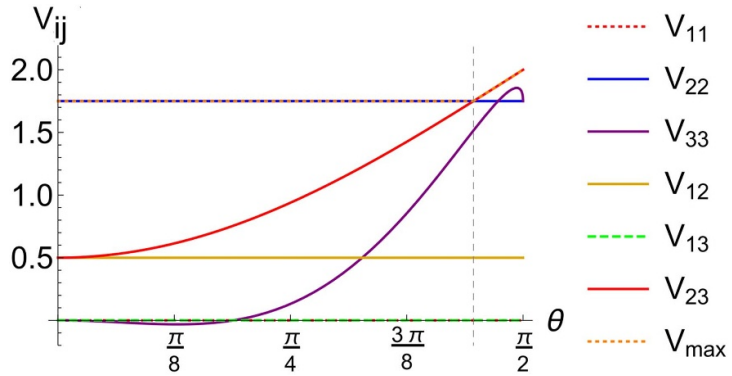

Figure 5. The variation of potential matrix elements with the driving parameter for our quantum modified PD game with $R = 6, S = 1, T = 7$, and $P = 5$. The black dashed line corresponds to the critical point, $\theta_c = 0.4467\pi$.

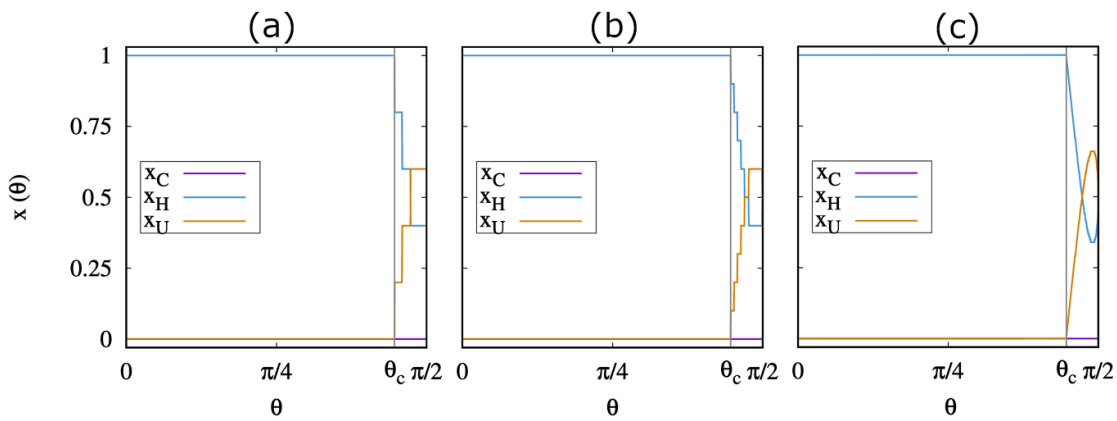

Figure 6. This figure exhibits the fractions of players corresponding to strategies C, H, and U(θ) for a quantum modified potential game. The underlying game is a quantum PD game with $R = 6, S = 1, T = 7$, and $P = 5$. Subplots (a), (b), and (c) correspond to all-to-all network with five ($N = 5$), ten ($N = 10$), and hundred ($N = 100$) players respectively.

One can see that V_{33} among the diagonal elements, and V_{13} & V_{23} among the off-diagonal elements are functions of θ. Obviously, the necessary condition for C to be NE is $V_{11} > V_{12}$ and $V_{11} > V_{13}$, i.e.

$$R > T \text{ and } 2R > T + P, \tag{21}$$

respectively. It implies that C is never a NE if the underlying game is PD, where $T > R > P > S$.

Similarly, for H to be most probable NE, the potential element V_{22} must be maximum. There exists an upper bound of θ (say, θ_c), beyond which V_{23} exceeds V_{22}. The expression for the upper bound of θ follows from the condition $V_{23} = V_{22}$:

$$\theta_c = 2\arcsin\sqrt{\frac{P + T - R - S}{4(2P - R - S)}}. \tag{22}$$

For a typical set of payoff elements $R = 6,\ S = 1,\ T = 7$, and $P = 5$, the critical value is $\theta_c \approx 0.4467\pi$. For this case, the variation of potential matrix elements as a function of the transition parameter and for maximal entanglement has been shown in figure 5. Henceforth, we consider the entanglement operator to be a perfect entangler ($\varepsilon = \pi/2$) for all the cases discussed in this paper.

For the regime below the critical point, the state of the system at zero temperature consists solely of players utilizing the strategy H. However, in the regime above the critical point, the population may exhibit a mixture of the three strategies: C, H, and U. This regime corresponds to the antiferromagnetic phase within the context of quantum potential games.

The maximum energy and the corresponding state of the population are determined by maximizing the total energy over all possible states. We present the state corresponding to the maximum energy in figure 6 for a PD game. It can be noted that this state only corresponds to a fraction of H and U for

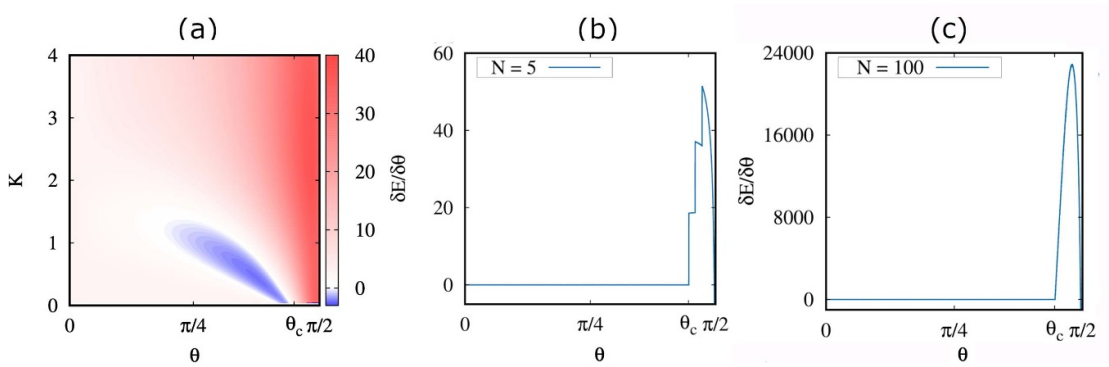

Figure 7. Illustrating quantum-like phase transitions in quantum PD game: the density plot (subplot (a) depict the first derivative of energy with respect to the transition parameter on parameter space of noise strength (K) and transition parameter (θ). We have considered an all-to-all connected network of five players and $R = 6, S = 1, T = 7$, and $P = 5$. The resemblance of this plot with quantum phase transition plots (see figure 1.2(b) in textbook [10]) is intriguing. Subplot (b) and (c) depict the first derivative of energy with respect to the transition parameter as a function of the transition parameter (θ) for $K = 0$ for $N = 5$ and $N = 100$ players, respectively.

$\theta > \theta_c$. From our numerical calculations, we observe that the strategy fractions (figure 6) tend to become continuous at the phase boundary as the system size increases.

Letting α to denote the number of players with strategy U at $\theta = \theta_c^+$, the expression for the energy at $\theta = \theta_c^+$ is given by the form:

$$\langle E \rangle \Big|_{\theta=\theta_c^+} = \binom{\alpha}{2} V_{33}(\theta_c) + \binom{N-\alpha}{2} V_{22} + \alpha (N-\alpha) V_{23}(\theta_c). \tag{23}$$

On the other hand, the energy at $\theta = \theta_c^-$ is independent of θ,

$$\langle E \rangle \Big|_{\theta=\theta_c^-} = \binom{N}{2} V_{22}. \tag{24}$$

Therefore, the difference between the first derivatives of the energies of the two phases at the phase boundary is given by the following expression:

$$\delta \left| \frac{\partial \langle E \rangle}{\partial \theta} \right| \equiv \left| \frac{\partial \langle E \rangle}{\partial \theta} \Big|_{\theta=\theta_c^+} - \frac{\partial \langle E \rangle}{\partial \theta} \Big|_{\theta=\theta_c^-} \right| = \left| \binom{\alpha}{2} \frac{\partial V_{33}(\theta_c)}{\partial \theta} + \alpha (N-\alpha) \frac{\partial V_{23}(\theta_c)}{\partial \theta} \right|. \tag{25}$$

Since both the potential elements V_{33} and V_{23} are smooth functions at the boundary, the discontinuity in the derivative of energy can appear if α is nonzero. Since from figure 6 we observe, as N increases, the value of α limits to zero at the phase boundary from right, i.e. at $\theta = \theta_c^+$, the value of the expression in equation (25) tends to zero in the thermodynamic limit. Here, we assume the system size N to be large but not infinity. From this insight, we can conclude that the derivative of energy (and hence the energy itself) with respect to θ in the thermodynamic limit are continuous functions at the phase boundary (see figures 7(b) and (c)). Thereby, it also brings forth a continuous ferromagnetic-to-antiferromagnetic phase transition in the context of quantum game.

Before proceeding further, we should point out that the entanglement operator employed herein is not the most general form. As an illustration, the operator used in [45, 59] differs from ours by a negative sign in the exponential term (see equation (17)). A straightforward calculation with this entanglement operator reveals that the corresponding payoff remains unchanged, apart from a relative phase factor in the final wavefunction. This implies that the physical results including the potentiality conditions (equation (19) and equation (C5)) are unaffected by this sign difference. More generally, the entanglement operator can be expressed in terms of a pair of Cartan parameters [60–62], of which the specific form used in our analysis arises as a special case. The analogues of equations (19) and (20) can likewise be derived under this generalized framework. However, for the class of 'perfect entanglers' (operators that generate maximum entanglement from initially unentangled qubits), as considered in our study, it can be shown that the final results remain invariant across all equivalent entanglement-operator formulations, apart from possible differences in the relative phase factors of the final wavefunction.

5. Hint of quantum-like phase transition

The resilience of the maximally probable NE near the zero temperature may be effectively characterized as the ratio of the potential difference (Δ) between the top two probable states and the minimum thermal fluctuation: Δ/K (recall that K is analogous to temperature in energy units). The higher this ratio the more resilient the rationality is with respect to thermal fluctuations. This gap in potentials between the top two most probable states is analogous to the energy gap of the lowest excitation above the ground state in quantum many-body systems [10]. At the phase transition, the system is extremely sensitive to the thermal fluctuations as this gap closes to zero. Even at an infinitesimal temperature, all possible strategic interactions are present in any many-body systems.

The theory of quantum many-body systems deals with the nature of phases of interacting quantum particles in their ground states. To delve into this, two different energy-scales are compared, namely, the Planck energy, representing the typical energy of the Hamiltonian, and the thermal energy. Consider a quantum Hamiltonian of a lattice-dependent model and a dimensionless coupling parameter in the system. For an infinite lattice, this energy of the ground state can no longer behave as a smooth, analytic function of this parameter. This situation occurs when an excited level shifts into the ground state and vice-versa causing a level-crossing at a specific value of the parameter. The signature of this transition is marked by the non-analyticity in the ground state energy. This kind of transitions are known as quantum phase transitions. Unlike classical phase transitions, here temperature does not govern the transition in the system.

The phase transition considered in this article is analogous to the above-mentioned quantum phase transition. Our system is a all-to-all coupled lattice (note each lattice point represents each individual of the population). The dimensionless parameter of phase transition depends on the system under study: for the classical game it is payoff elements whereas for the quantum game it is the strategy parameter. In line with expectation, we also observed discontinuous shift in internal energy at the critical point.

In the study of quantum phase transitions, researchers utilize a concept known as fidelity, which serves as an information-theoretic measure to identify the ground state singularities linked to a quantum phase transition. Fidelity is quantified as the overlap between the ground states of a specific Hamiltonian evaluated at two different settings of the tuneable parameter (say, λ). Formally, fidelity for two ground states $|\psi_0(\lambda)\rangle$ and $|\psi_0(\lambda+\delta\lambda)\rangle$ of a Hamiltonian $H(\lambda)$ is defined as the overlap of states very close to each other in the parameter space [63]:

$$\mathcal{F}(\lambda) = \lim_{\delta\lambda\to 0} |\langle\psi_0(\lambda)|\psi_0(\lambda+\delta\lambda)\rangle|. \tag{26}$$

The above quantity is bounded from above by unity (corresponds to the case where the two states are identical) and from below by zero (corresponds to the case where the two states are orthogonal). The fidelity, $\mathcal{F}(\lambda)$, vanishes in the limit of the states becoming mutually orthogonal: it is generally the case close to a quantum critical point [10, 63]. It is observed that the fidelity shows a sharp decay in proximity to a critical point indicating a signature of phase transition [63–65].

To assess the population surrounding the critical point, in line with our agenda to digging up analogies with quantum phase transitions, we require a comparable metric. Using the transition parameter, λ, we define a fidelity-like measure for our analysis as follows:

$$f(\lambda) = 1 - \frac{d(\lambda)}{\sqrt{2}}, \tag{27}$$

where $d(\lambda)$ denotes the normalized Euclidean distance between the population states corresponding to the parameters λ and $\lambda+\delta\lambda$ ($\delta\lambda\to 0$). Any discontinuity observed in $f(\lambda)$ would highlight the critical point in the game-theoretic population akin to the behavior seen in quantum phase transitions within many-body systems.

Let $(x_{s_1}(\lambda), x_{s_2}(\lambda), \cdots, x_{s_n}(\lambda))$ represent the population state, where the fraction $x_{s_i}(\lambda)$ denotes the fraction of the population equipped with the strategy s_i at a given λ. The formula for $d(\lambda)$ can then be expressed as follows:

$$d(\lambda) = \lim_{\delta\lambda\to 0} \sqrt{\sum_{i=1}^{n} [x_{s_i}(\lambda+\delta\lambda) - x_{s_i}(\lambda)]^2}. \tag{28}$$

The range of $d(\lambda)$ is $[0, \sqrt{2}]$. It is worth considering the relevance of fidelity-like measures in game-theoretic contexts. Such measures, originally introduced in quantum and statistical physics to quantify

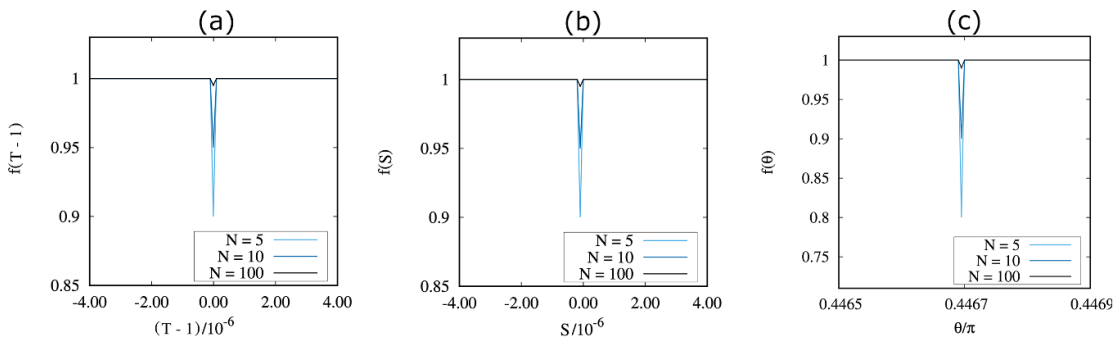

Figure 8. Quantum-like phase transitions captured by fidelity-like measure. (a) and (b) correspond to the transitions between phases I and II, and between phases II and III respectively in 2×2 classical game (refer equation (7)). (c) corresponds to the phase transition in quantum PD game ($R = 6, S = 1, T = 7$, and $P = 5$; refer section 4). We have taken $\delta\lambda = 10^{-7}$ and $\delta\theta = \pi/10^{-6}$ in the classical and the quantum games, respectively.

the similarity between states under parametric variations, can offer valuable insight into the stability and sensitivity of equilibria in evolutionary games. From the game-theoretic perspective, $d(\lambda)$ denotes the distance in the state space—represented by an $(n-1)$-dimensional simplex—between two equilibrium population states corresponding to infinitesimally close parameter values λ and $\lambda + \delta\lambda$. By construction, an increase in this distance implies a decrease in the associated fidelity, and vice versa. In particular, a fidelity value of unity ($f = 1$) indicates that the equilibrium (or ground) state remains invariant with respect to small changes in the control parameter, reflecting local stability of the population configuration. Conversely, a fidelity approaching zero ($f \to 0$) signifies that the two equilibrium states become orthogonal, i.e. there is a complete reorganization of the population state distribution. This sharp drop in fidelity can thus serve as a precursor or diagnostic indicator of a qualitative change in the behavior of the system—analogous to a quantum or thermodynamic phase transition. In this sense, fidelity provides a unified, geometry-based metric to quantify how sensitively an equilibrium responds to perturbations in the underlying game parameters, bridging microscopic variations in payoffs with macroscopic structural changes in the population dynamics.

It is clear that the phase transition between ferromagnetic phases retains a fidelity of 1, except at the critical point, where the fidelity drops to zero. In contrast, during the transitions between antiferromagnetic and ferromagnetic phases, the fidelity varies continuously across the transition point in the thermodynamic limit. However, for finite populations, there is a discontinuity of $1/(2N)$ across the phase boundaries. For more details, please refer to appendix D. For the case of continuous phase transition in the quantum game, we plot the corresponding fidelity in figure 8(c) which substantiates our claim of the phase transition.

6. Discussion and conclusions

This paper extends the efforts of interdisciplinary researchers in finding parallels between physics of phase transitions and transitions in population states due to modifications in parameters of game played between its individuals: although classical phase transitions already had known analogues in game theory [9, 66–68], curiously, the analogue of the quantum phase transition was missing in the game theory literature. In this paper, we comprehensively filled this lacuna using examples of both classical and quantum games.

In a typical game theoretic setup, there is a trade off between the potential difference between the energy levels (a function of the payoff matrix elements) and the noise parameter. If the players are so rational that the dynamics is solely governed by the potential elements, the quantum-like phase transition occurs. This phenomenon is similar to the case of quantum phase transition where the characteristic energy scale of the thermal fluctuations above the ground state vanishes.

We have presented both continuous and discontinuous phase transitions, and also proposed an analogue of fidelity measure used in quantum phase transitions in physics problems. One important observation is that just alteration of NE is not enough for the kind of phase transitions we have discussed in this paper; rather, the statistical mechanical framework requires alteration of most probable NE—something that can be located using potential matrix formulation—to result in the aforesaid phase transitions.

While ground-state fidelity has been generally used to detect quantum phase transitions [63], it often fails to serve as a reliable marker, especially for systems with all-to-all connectivity, like the Lipkin–Meshkov–Glick model [64, 69]. Generally in these systems, the phase transitions are driven by collective, mean-field-like behavior. As a result, the overlap between ground states does not show a sharp signature near critical points, especially in the thermodynamic limit. Numerical studies also reveal that fidelity minima are often broad or absent due to finite-size effects, making the critical point difficult to identify. Although fidelity shows promise in some short-range, a clear and general criterion for when it acts as a proper signature for a transition is still lacking [65]. In view of these results in literature of quantum many-body systems, one should also generalize our approach on systems where agents interact on short-ranged graphs to observe a closer similarity to quantum phase transitions in short-ranged ordered lattices.

In the framework discussed in this paper, it is interesting to realize that entanglement effectively functions as a shared non-classical correlation that facilitates coordinated strategic decision-making among agents without the need for explicit communication. This correspondence between quantum correlations and economic coordination mechanisms underscores the possibility that quantum game-theoretic models may naturally arise in financial and economic systems, particularly in environments where strategic agents interact under conditions of limited or asymmetric information [70, 71].

The alteration of NE with the variation of game-theoretic parameters is not, by itself, a new observation [72, 73]; however, what is distinctive in our potential-based formulation lies in the physical interpretation it affords: firstly, we embed potential games within the conceptual and mathematical framework of *statistical physics*, thereby establishing a formal correspondence between microscopic interactions (individual strategic behaviors) and macroscopic observables (aggregate system states). This mapping allows us to treat the collective behavior of strategic agents analogously to that of particles in a thermodynamic ensemble, where the system tends toward configurations that maximize a scalar potential analogous to free energy. Secondly, within this framework, a mere change in the Nash equilibrium does not automatically imply a *phase transition*. Instead, a true transition occurs only when the *maximum-potential state*—the strategic configuration that globally maximizes the potential—undergoes a switch. This phenomenon directly parallels a *ground-state transition* in a many-body physical system, where a qualitative change in the dominant configuration marks a new phase. To further illustrate this correspondence, we explicitly note:

1. the *temperature* of the physical system maps onto the *irrationality* or noise parameter in game theory, quantifying the degree of stochasticity or deviation from perfectly rational behavior in the population;
2. the *average potential* of the game corresponds to the *average energy* (or free energy) of the physical system, representing the expected payoff at the population level; and
3. a *flip in the maximum-potential state* mirrors a *ground-state transition*, signifying a macroscopic reorganization of the system and, necessarily, a shift in the NE.

This analogy not only clarifies the nature of equilibrium transitions in evolutionary games but also provides a principled way to classify them using tools and intuitions drawn from statistical mechanics.

The equilibrium population states of the system, as studied herein, can be obtained as the end results of Logit rule [74–76] (equivalent to Glauber dynamics in non-equilibrium statistical mechanics [8, 77, 78]). Thus, an immediate research avenue worth pursuing as an extension of our work is to investigate analogies between dynamical quantum phases [10, 63] and the non-equilibrium states of population consisting of players playing strategic games. We envisage that such an equivalence would help in novel characterization of the learning processes of agents though the concepts and measures used for characterizing various aspects of dynamical quantum phase transitions [79–81].

While our formulation and results retain complete generality and applicability for the class of two-player–two-strategy games, as the strategy space increases-leading to higher-dimensional games-the situation becomes more restrictive. In such cases, the existence of a potential function is no longer guaranteed. Payoff structure with a well-defined potential naturally becomes a necessary condition for extending our framework to these more complex systems. This distinction bears a close analogy with classical physics: the dynamics of conservative force fields can be described within the framework of equilibrium statistical mechanics, where a scalar potential (energy function) fully determines the system's evolution and stationary states. In contrast, non-conservative or dissipative force fields lack such a potential representation, and hence, no general statistical-mechanical formulation can be constructed for them. Similarly, non-potential games do not admit a global potential, making them incompatible with the equilibrium-based analytical tools employed in our study.

A potential structure is not guaranteed even in quantum games involving more than three strategies. Even within the framework of a three-strategy quantum game-particularly when two of the strategies do not possess any classical analog-the system exhibits highly complex and nonlinear behavior. In such cases, it becomes extremely difficult to predict the overall outcome if one replaces the Hadamard strategy with any other quantum strategy. This difficulty arises due to the pronounced nonlinearities embedded in the payoff landscape, which can lead to significant shifts in the phase transition points, alterations in the corresponding sets of NE, and even changes in the nature of the phase transitions. Consequently, the analytical characterization of such higher-dimensional or more generalized quantum strategic spaces remains a formidable challenge. The system may exhibit entirely new regimes depending on the choice of entanglement parameters and the specific form of the quantum strategies employed. Therefore, any comprehensive or general statement about these extended scenarios would require a separate and detailed investigation, which lies beyond the current scope of this work but represents an interesting direction for future research.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

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Appendix A. Second highest energy configuration

Let us assume that V_{mm} and $V_{m'm''}$ be the largest and the second largest elements of the potential matrix. We consider following two cases separately:

Case 1: $m' = m''$. The system, in its second maximum energy state, in principle, can have α players playing the strategy m and the rest $(N-\alpha)$ playing the strategy m'. The value of α ranges between $1 \leqslant \alpha \leqslant N$. We now prove that $\alpha = 1$ corresponds to the second maximum energy state in the thermodynamic limit. Let $E(N_1, N_2)$ denote the energy of the state where N_1 and N_2 agents are playing the strategies m and m', respectively. Let us compare the energies of the states corresponding to α and $(\alpha+1)$. Mathematically, lets first examine the case for $\alpha = 0$:

$$E(N,0) - E(N-1,1) = (N-1)(V_{mm} - V_{mm'}) > 0. \tag{A1}$$

Now we move to the generic case,

$$\begin{aligned} E(N-\alpha,\alpha) - E(N-\alpha-1,\alpha+1) &= \binom{N-\alpha}{2} V_{mm} + \binom{\alpha}{2} V_{m'm'} + \alpha(N-\alpha) V_{mm'} \\ &\quad - \binom{N-\alpha-1}{2} V_{mm} + \binom{\alpha+1}{2} V_{m'm'} + (\alpha+1)(N-\alpha-1) V_{mm'} \\ &= (N-\alpha-1) V_{mm} - (N-2\alpha-1) V_{mm'} > 0 \quad \forall \alpha. \end{aligned} \tag{A2}$$

Therefore, by mathematical induction, it is evident that $\alpha = 1$ corresponds to the second maximum energy state in the thermodynamic limit.

Case 2: $m' \neq m''$. The system, in its second maximum energy state, in principle, can have α players playing the strategy m', β players playing the strategy m'', and the rest $(N-\alpha-\beta)$ playing the strategy m. The values of α and β range between $1 \leqslant \alpha \leqslant N$. Let $E(N_1, N_2, N_3)$ denote the energy of the state where N_1, N_2, and N_3 agents are playing the strategies m, m', and m'', respectively. We prove now that the case $\alpha = 1$ corresponds to the second maximum energy state in the thermodynamic limit. Following the same footsteps as used for the last case, we arrive at the following inequality:

$$E(N-\alpha-\beta,\alpha,\beta) - E(N-\alpha-\beta-1,\alpha+1,\beta) = (N-\alpha-\beta-1)(V_{mm} - V_{mm'}) > 0 \quad \forall \alpha, \beta. \tag{A3}$$

From the above inequality, it can be concluded by mathematical induction that $\alpha = 1$ and $\beta = 0$ (or, $\alpha = 0$ and $\beta = 1$, depending on the scenario) corresponds to the second maximum energy state in the thermodynamic limit, i.e. the population state corresponding to the second highest energy configuration is always the case where $(N-1)$ players play the strategy m and only one plays m' (or, m'').

Appendix B. Ferromagnetic to ferromagnetic transition: a case study

For a symmetric 2×2 classical game, consider a path in the (S, T)–plane that begins in the region $1 > T > S > 0$ (refer to equation (7) and the subsequent discussion), passes exactly through the degenerate point $(S = 0, T = 1)$, and then continues into the region $T > 1 > 0 > S$. The degenerate point acts as a boundary between two game classes: the Harmony game (HG) and the Prisoner's Dilemma (PD) respectively. Before reaching the degenerate point, the game has a unique symmetric pure NE (C, C) because $T < 1$ and $S > 0$. At the degenerate point, a player receives the same payoff from choosing C or D, regardless of the opponent's action. Consequently, every pure profile and every mixed profile is a Nash equilibrium at this point.

Immediately after passing the point into the PD region-$T > 1$, $S < 0$ (for example, $S = -0.5$, $T = 1.2$)-the cooperative equilibrium is destroyed while the defection equilibrium appears, leaving (D, D) as the unique symmetric pure NE. Thus, when the path crosses exactly through the degenerate point, the equilibrium structure changes discontinuously. In practice, this means that at the moment of crossing, the players are indifferent, and arbitrarily small perturbations determine the equilibrium around the degenerate point. In a population of N players, the equilibrium state therefore switches from all cooperation to all defection as the payoff parameters cross the degenerate point.

The same phenomenon can be interpreted using the potential function within the framework of equilibrium statistical mechanics. The Hamiltonian in equation (2), corresponding to a many-agent all-to-all interaction, can be expressed in terms of the strategy variables s^{μ} and s^{ν}, which take the values 1 (for strategy C) and 2 (for strategy D), for a symmetric 2×2 classical game. The elements of the global potential matrix in this case are computed using equation (1) as

$$V_{11} - V_{21} = 1 - T,$$
$$V_{22} - V_{12} = -S.$$

Owing to the gauge invariance of energy, one of the four elements can be set to zero without loss of generality. We therefore set $V_{11} = 0$ to obtain equation (8).

In a population of N players, each can choose between two strategies, C or D, resulting in 2^N possible configurations. If one of the diagonal elements of the potential matrix-say V_{mm} $(m \in \{1, 2\})$-attains the maximum value, then under the assumption of perfect rationality, all players will choose the strategy corresponding to V_{mm}, leading to the *state of maximum potential*. However, in the absence of perfect rationality, other configurations with lower potential values may also coexist in the system. These configurations are captured by the terms appearing in the partition function (see equations (3) and (5)). The partition function represents the total measure of how all possible strategy profiles are weighted according to their potential values at a given finite rationality. Intuitively, it quantifies the likelihood of the system settling into high-potential (more favorable) configurations as compared to low-potential ones.

For example, a configuration with the second-highest potential may consist of $(N-1)$ players interacting through V_{mm}, while one player deviates and interacts with the rest via the second-highest potential element $V_{mm'}$ $(m, m' \in \{1, 2\})$. Therefore, in the limit of small noise (corresponding to almost rational players), the partition function and the internal energy can be expanded as discussed in equations (5) and (6).

To illustrate the transition between the two ordered collective states (a ferromagnetic-to-ferromagnetic transition), consider two sets of payoff parameters: one corresponding to the HG and the other to the PD.

HG: For $T = 0.7$ and $S = 0.5$, the potential matrix becomes

$$\mathsf{V} = \begin{pmatrix} 0 & -0.3 \\ -0.3 & -0.8 \end{pmatrix}.$$

The largest potential element is $V_{11} = 0$, corresponding to mutual cooperation. Hence, when all players choose C, the total potential is maximized. Under perfect rationality, this fully cooperative configuration

forms the stable equilibrium (NE) of the Harmony Game. In this scenario, in the presence of small noise, using equations (3) and (5), the partition function can be written explicitly as

$$Z(K) = 1 + N\exp\left[-\frac{0.3\,(N-1)}{K}\right]. \tag{B1}$$

Similarly, from equation (6), the internal energy can be expressed as

$$\langle E\rangle = -\frac{0.3N(N-1)}{N+\exp\left[\frac{0.3(N-1)}{K}\right]}. \tag{B2}$$

Obviously, for perfect rationality ($K=0$), the total energy is

$$\langle E\rangle\,|_{K=0} = 0. \tag{B3}$$

PD: For $T=1.2$ and $S=-0.5$, the potential matrix becomes

$$\mathsf{V} = \begin{pmatrix} 0 & 0.2 \\ 0.2 & 0.7 \end{pmatrix}.$$

Here, the largest potential element is $V_{22}=0.7$, corresponding to mutual defection. Therefore, when all players choose D, the total potential is maximized. This configuration represents the Nash equilibrium of the PD. Here, in the presence of small noise, from equations (3) and (5), the partition function becomes

$$Z(K) = \exp\left[\binom{N}{2}\times 0.7\right]\left(1+N\exp\left[-\frac{0.5\,(N-1)}{K}\right]\right), \tag{B4}$$

and the corresponding internal energy is given by

$$\langle E\rangle = 0.7\binom{N}{2} - \frac{0.5N(N-1)}{N+\exp\left[\frac{0.5(N-1)}{K}\right]}. \tag{B5}$$

Therefore, under perfect rationality, the energy of the system is

$$\langle E\rangle\,|_{K=0} = 0.7\binom{N}{2}. \tag{B6}$$

The transition from global cooperation to global defection as the parameters (T,S) are varied thus represents a *ferromagnetic-to-ferromagnetic transition* between two ordered collective states, each characterized by full alignment of the players' strategies and change in total energy of the system (zero versus non-zero—as seen above in equations (B3) and (B6)). The point $(S=0,\,T=1)$ on the line $T-S-1=0$ serves as the boundary connecting the two game classes—HG and PD—summarized in table 1.

Appendix C. Three–strategy quantum potential game

A two-player three-strategy quantum game essentially allows the players to choose at least one pure quantum strategy. By the word, *pure*, we denote all such quantum strategies that are not accessible in a classical game setup. One knows that for a symmetric three-strategy classical game to be a potential game, the incentive of pairwise inter-strategy interactions is symmetric with respect to strategy-interchange among the players [3]. Thus, the condition of potentiality for such three-strategy symmetric games ($\Pi^A=\Pi^B=\Pi$) turns out to be as follows:

$$\sum_{i=1}^{3}\Pi\left(s_i,s_{i+1}\right) = \sum_{i=1}^{3}\Pi\left(s_{i+1},s_i\right), \tag{C1}$$

with $s_4\equiv s_1$.

This condition of potentiality should be sought for the quantum potential game. For such a game, with three strategies s_1, s_2 and s_3 more appropriately denoted as U_1, U_2 and U_3, respectively, the potential the condition can be recast as:

$$\sum_i \Pi\left(\mathsf{U}_i,\mathsf{U}_{i+1}\right) = \sum_i \Pi\left(\mathsf{U}_{i+1},\mathsf{U}_i\right) \tag{C2a}$$

$$(S-T)\left[\sum_i |\psi_{01}(\mathsf{U}_i,\mathsf{U}_{i+1})|^2 - \sum_i |\psi_{10}(\mathsf{U}_i,\mathsf{U}_{i+1})|^2\right] = 0, \tag{C2b}$$

The first equation is just a rewriting of equation (C1), whereas the second equation follows from the fact that $\mathsf{J}(\epsilon)$ is symmetric with respect to both the qubits.

The output state $|\psi_{\text{final}}\rangle = \mathsf{J}^\dagger(\varepsilon)(\mathsf{U}^A \otimes \mathsf{U}^B)\mathsf{J}(\varepsilon)|00\rangle$ of the quantum game is described by the following wavefunctions $\psi_{ij}(\mathsf{U}_A,\mathsf{U}_B) = \langle ij|\psi_{\text{final}}\rangle$:

$$\psi_{00}(\mathsf{U}_A,\mathsf{U}_B) = \cos^2\frac{\varepsilon}{2}\gamma_A\gamma_B + \sin^2\frac{\varepsilon}{2}\gamma_A^*\gamma_B^*, \tag{C3a}$$

$$\begin{aligned}\psi_{01}(\mathsf{U}_A,\mathsf{U}_B) = &-\cos^2\frac{\varepsilon}{2}\gamma_A\delta_B - \sin^2\frac{\varepsilon}{2}\gamma_A^*\delta_B \\ &+ i\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\delta_A\gamma_B^* - \delta_A\gamma_B),\end{aligned} \tag{C3b}$$

$$\begin{aligned}\psi_{10}(\mathsf{U}_A,\mathsf{U}_B) = &-\cos^2\frac{\varepsilon}{2}\delta_A\gamma_B - \sin^2\frac{\varepsilon}{2}\delta_A\gamma_B^* \\ &+ i\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\gamma_A^*\delta_B - \gamma_A\delta_B),\end{aligned} \tag{C3c}$$

$$\psi_{11}(\mathsf{U}_A,\mathsf{U}_B) = \delta_A\delta_B - i\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\gamma_A\gamma_B - \gamma_A^*\gamma_B^*). \tag{C3d}$$

We consider a game where both the players have access to a pure quantum strategy (say, $\mathsf{U} \equiv (\gamma(\theta,\phi),\delta(\theta))$) along with the classical strategies, Cooperation (C) and Defection (D). In this case,

$$\psi_{01}(\mathsf{C},\mathsf{D}) = 1, \tag{C4a}$$

$$\psi_{10}(\mathsf{C},\mathsf{D}) = 0, \tag{C4b}$$

$$\psi_{01}(\mathsf{D},\mathsf{U}) = i\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\gamma^* - \gamma), \tag{C4c}$$

$$\psi_{10}(\mathsf{D},\mathsf{U}) = -\cos^2\frac{\varepsilon}{2}\gamma - \sin^2\frac{\varepsilon}{2}\gamma^*, \tag{C4d}$$

$$\psi_{01}(\mathsf{U},\mathsf{C}) = 0, \tag{C4e}$$

$$\psi_{10}(\mathsf{U},\mathsf{C}) = -\delta. \tag{C4f}$$

Using equation (C4) in equation (C2), we find the following condition for potentiality to hold:

$$\begin{aligned}&[\Re(\gamma)]^2 + [\Im(\gamma)]^2\cos 2\varepsilon + \delta^2 = 1 \\ \Longrightarrow\ &\sin\varepsilon\cdot\Im(\gamma) = 0.\end{aligned} \tag{C5}$$

This essentially implies that unless the initial state is unentangled or the quantum strategy is in $\phi = 0$ subspace, the game is not potential. This is along the expected line as in the limit of zero entanglement the payoffs can be uniquely mapped to a payoff corresponding to a classical mixed strategy, and such a game is always potential.

Now we consider a game where both the players have assess to one classical strategy C, and two quantum strategies: Hadamard gate H and $\mathsf{U} \equiv (\gamma(\theta,\phi),\delta(\theta))$. In this case,

$$\psi_{01}(\mathsf{C},\mathsf{H}) = -\frac{1}{\sqrt{2}} \tag{C6a}$$

$$\psi_{10}(\mathsf{C},\mathsf{H}) = 0,$$

$$\begin{aligned}\psi_{01}(\mathsf{H},\mathsf{U}) = &-\frac{\delta^*}{\sqrt{2}}\cos^2\frac{\varepsilon}{2} - \frac{\delta}{\sqrt{2}}\sin^2\frac{\varepsilon}{2} \\ &+\frac{i}{\sqrt{2}}\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\gamma^* - \gamma),\end{aligned} \tag{C6b}$$

$$\begin{aligned}\psi_{10}(\mathsf{H},\mathsf{U}) = &-\frac{\gamma}{\sqrt{2}}\cos^2\frac{\varepsilon}{2} - \frac{\gamma^*}{\sqrt{2}}\sin^2\frac{\varepsilon}{2} \\ &+\frac{i}{\sqrt{2}}\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\delta - \delta^*),\end{aligned} \tag{C6c}$$

$$\psi_{01}(\mathsf{U},\mathsf{C}) = i\cos\frac{\varepsilon}{2}\sin\frac{\varepsilon}{2}(\delta - \delta^*), \tag{C6d}$$

$$\psi_{10}(\mathsf{U},\mathsf{C}) = -\delta^*\cos^2\frac{\varepsilon}{2} - \delta\sin^2\frac{\varepsilon}{2}. \tag{C6e}$$

Using equation (C6) in equation (C2), we find the following condition for potentiality to hold:

$$\begin{aligned}&[\Re(\gamma)]^2+[\Im(\gamma)]^2\cos 2\varepsilon+[\Im(\gamma)]\,\delta\sin\varepsilon+\delta^2=1\\&\implies \sin\varepsilon\sin\phi=\tan\frac{\theta}{2}.\end{aligned}\tag{C7}$$

This is essentially equation (19) used in the main text. It means that in the set-up of two-player game where each player's strategy set is $\{\mathsf{C},\mathsf{H},\mathsf{U}(\theta,\phi)\}$, the game will have a potential structure only if the parameters in the third strategy $\mathsf{U}(\theta,\phi)$ satisfy this transcendental equation. For the case of maximum entanglement ($\varepsilon=\pi/2$), the equation simply leads to $\sin\phi=\tan\frac{\theta}{2}$ and thereby restricting $\theta\in[0,\pi/2]$ (recall, $\phi\in[0,\pi/2]$).

Appendix D. Fidelities

Although a straightforward matter, for the sake of completeness, we present here calculations of fidelities as reported in table 1. In phase II, the equilibrium cooperator fraction is given by equation (14) where we put $T=1$ to get $x_C=1-1/(2N)$. Therefore, from equation (27) and equation (28), we arrive at

$$\begin{aligned}f(\lambda)\Big|_{T=1}&=1-\frac{\sqrt{\left[1-\left(1-\frac{1}{2N}\right)\right]^2+\left[0-\frac{1}{2N}\right]^2}}{\sqrt{2}}\\&=1-\frac{1}{2N}.\end{aligned}\tag{D1}$$

which is continuous in the thermodynamic limit as it approaches 1—the value of fidelity deep inside any single phase.

Similarly, the fidelity between phases II and III, is found by setting $S=0$ in equation (14) to find $x_C=1/(2N)$ leading to

$$\begin{aligned}f(\lambda)\Big|_{S=0}&=1-\frac{\sqrt{\left[0-\frac{1}{2N}\right]^2+\left[1-\left(1-\frac{1}{2N}\right)\right]^2}}{\sqrt{2}}\\&=1-\frac{1}{2N}.\end{aligned}\tag{D2}$$

Lastly, the fidelity between phases I and III, is

$$f(\lambda)\Big|_{S=T-1}=1-\frac{\sqrt{(1-0)^2+(1-0)^2}}{\sqrt{2}}=0\tag{D3}$$

which is discontinuous at the boundary even as $N\to\infty$.

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